

GapTwin

A Sensor-Optional Digital Twin for Assembly Line Bottleneck & Defect Prediction

Business Proposal — Round 2

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1. Problem Framing

The problem isn't that plants lack data. The problem is that the data is uneven, fragmented, and often arrives too late to prevent the problem.

On a vehicle assembly line, no station operates in isolation. A slowdown, abnormal process condition, or quality issue at one station can propagate through everything downstream. A small increase in cycle time can become a growing queue, reduced throughput, delayed vehicles, and eventually a much larger production loss. In parallel, an early process deviation may continue across multiple vehicles before the defect is discovered at end-of-line inspection.

Today, plant teams are often forced to react to consequences rather than act on early signals. This is not a lack of effort; it is a visibility problem. Real assembly lines are a mix of modern, well-instrumented equipment and older or manually operated stations. Some stations provide rich sensor data, while others provide little or no direct measurement. A useful digital twin therefore cannot assume that every part of the line is equally observable.

The core challenge is to provide plant teams with a timely, line-wide understanding of what is happening, where a deviation is forming, how serious it is, what it could affect downstream, and what should be addressed next — even when sensor coverage is incomplete.

2. Solution Design

2.1 Overview

GapTwin is a live operational digital twin of the assembly line. It represents the state and behaviour of stations, buffers, vehicle flow, and quality signals without trying to recreate every robot, component, or physical interaction in a 3D environment. The focus is decision-making: turning production data into a clear view of current conditions, emerging risks, likely impact, and possible interventions.

GapTwin continuously assesses station health and maintains a lightweight, per-vehicle quality-risk profile. This creates two connected views of the same production process: the health of the line and the quality exposure of each vehicle moving through it.

The defining principle is sensor-optional visibility. Instrumented stations can be assessed directly from their available signals. At stations with limited or no direct sensor coverage, the system can infer abnormal behaviour from timing patterns, buffer buildup, starvation, and the behaviour of nearby instrumented stations. In other words, no sensor does not mean no visibility.

2.2 Core Mechanism

Station health monitoring: Track cycle time against each station's expected/takt time and compare recent behaviour with its own historical baseline. Sustained deviation is treated differently from a single noisy reading, helping reduce false alarms.

Sensor-optional inference: For stations without direct instrumentation, infer abnormal behaviour from the timing relationship between neighbouring instrumented stations, including elapsed-time drift, buffer growth, and starvation patterns.

Confidence-scored alerts: Every alert states whether the evidence is directly measured or inferred and communicates the confidence of the assessment. Low-confidence inference is presented as a probable area, not a certainty.

Propagation / blast-radius analysis: When a station begins degrading, estimate the downstream effect: queue growth, affected stations, potential throughput loss, and the number of vehicles exposed during the relevant time window.

Predicted lead time: Instead of only saying that a station is abnormal, estimate the runway before a critical threshold is likely to be crossed when enough data exists. For example: "At the current drift rate, the station may cross the critical threshold in ~18 minutes."

Vehicle Quality Health: Maintain a per-vehicle quality-risk profile based on the conditions the vehicle experienced at earlier stations. If a station is degraded during a defined exposure window, the system can identify which vehicles passed through that window for targeted inspection.

Defect genealogy: When a defect is identified, trace backward through the vehicle's exposure history and historical patterns to highlight the likely station or process point associated with the deviation. The system does not claim to diagnose the physical root cause.

Parameter optimization: For a confirmed or emerging bottleneck, evaluate historical operating conditions to identify parameter combinations associated with better cycle time and throughput while keeping quality risk and operating limits within acceptable boundaries. Candidate configurations are evaluated before the plant team decides whether to act.

Human-in-the-loop: GapTwin observes, prioritizes, predicts, and recommends. Plant personnel remain responsible for physical inspection, parameter changes, maintenance decisions, and production actions.

2.3 Decision Loop

GapTwin is designed around a simple operational loop:

Detect → Understand the blast radius → Identify affected vehicles → Estimate time to impact → Find the best safe operating configuration → Let the plant team decide.

2.4 Alert Prioritization

Plant teams should not receive a wall of equal-priority warnings. Alerts can be ranked using a combination of severity, downstream impact, confidence, and time-to-threshold. This helps the floor supervisor focus first on the issue that is most likely to create meaningful production or quality loss.

2.5 Deliberate Non-Scope

- No full physical assembly simulation: GapTwin does not attempt detailed robotics, CAD, physics, or component-level simulation.
- No unsupported root-cause claims: the system identifies where a deviation is most likely originating and with what confidence; physical diagnosis remains with engineering and plant teams.

- No autonomous machine control: GapTwin recommends and simulates options but does not directly change machine settings.
- No requirement for full sensor coverage: the initial value proposition is specifically designed around mixed instrumentation.

3. Target Users & Decisions

User	Primary Need	GapTwin View	Decision Supported
Floor Supervisor	Immediate operational awareness	Station status, ranked alerts, lead time, blast radius, next actions to consider	What needs attention first?
Plant Manager	Planning and performance visibility	Trends, bottleneck patterns, avoided-loss estimates, rollout progress	Where should resources and improvement effort go?
Quality / QC	Early, targeted quality intervention	Vehicle quality-risk score and traceable exposure history	Which vehicles should be inspected first?
Maintenance / Engineering	Investigation and intervention support	Station degradation history, parameter context, likely origin area, what-if configurations	What should be checked or tested?
Plant Leadership	Business case and scale	ROI indicators, phased deployment roadmap, cross-line scalability	Where does the solution create enough value to scale?

4. Business Case & Impact

Reduced rework: Earlier detection limits the number of vehicles produced under degraded conditions and reduces the chance of a problem compounding across a batch.

Targeted inspection: Instead of treating an entire batch as equally risky, quality teams can focus on vehicles that actually passed through a degraded station during the relevant exposure window.

Lower retrofit dependency: A pilot can start with the sensor coverage that already exists. The system can infer visibility where direct measurement is missing, avoiding the need to wait for a large instrumentation project.

Faster intervention: Lead-time estimates give teams a warning window instead of a binary alarm, creating more opportunity to inspect, adjust, or plan a controlled response.

Better use of engineering effort: Parameter optimization and what-if analysis can narrow the set of operating configurations worth testing instead of relying only on trial and error.

Compounding plant value: The same monitoring and inference framework can be applied across mixed sensor environments and expanded line by line.

4.1 Example: Exposure Window

Suppose Station S4 shows sustained degradation from 10:20 to 10:40. If 17 vehicles passed through S4 during that period, GapTwin can report “17 vehicles potentially exposed” and show their subsequent process history. It should not claim that all 17 vehicles are defective. This distinction keeps the system useful without overstating what the data proves.

4.2 Example: Bottleneck Optimization

If S4 is predicted to become the bottleneck, GapTwin can compare historical operating conditions for S4 under similar production contexts. It can surface a small set of candidate parameter configurations that historically delivered lower cycle time without an unacceptable increase in defect risk, then present those options to the plant team for validation before any change is made.

5. Rollout Roadmap

1. Pilot: Single line, existing sensor coverage, no retrofit. Validate detection, inference, lead-time estimates, and exposure tracing against historical incidents.

2. Augment: Add low-cost camera or other lightweight sensing at the highest-value unsensored stations where additional evidence would materially improve confidence.

3. Expand: Extend to additional lines during scheduled maintenance windows, minimizing disruption to live production.

4. Scale: Build a cross-plant pattern library so validated detection and intervention patterns can transfer faster to similar lines and processes.

6. Risks & Mitigations

False alarms: Use sustained-deviation thresholds, station-specific baselines, confidence scoring, and alert prioritization rather than reacting to every abnormal reading.

Inference points to the wrong station: Clearly distinguish inferred from directly measured conditions, lower confidence when evidence is weak, and present a probable area rather than false certainty.

Lead-time estimates are unreliable: Require a minimum data sufficiency bar. If the trend is too early, noisy, or slow-moving to support a credible estimate, show no lead-time number rather than inventing one.

Workforce surveillance concerns: Attribute conditions to stations and process windows, not individual operators. The goal is process improvement, not employee scoring.

Plant-to-plant variation: Use configurable station baselines, takt assumptions, operating limits, vehicle variants, and local process context.

Optimization suggests an unsafe or unsuitable setting: Treat optimization as constrained decision support. Candidate configurations must remain within predefined operating and quality limits and require human validation before use.

7. Why GapTwin Is Different

Many digital twin concepts focus on highly instrumented factories or detailed physical simulation. GapTwin takes a different approach: it is designed around the reality of mixed instrumentation and the decisions plant teams actually need to make.

- Sensor-optional by design — incomplete instrumentation does not automatically mean an invisible station.
- Propagation-aware — the system looks beyond the local alert to understand the downstream blast radius.
- Vehicle-level quality visibility — risk can be connected to the specific units exposed to abnormal conditions.
- Time-aware — alerts can include an estimate of how much runway remains before a critical threshold is reached.

- Optimization-aware — once a bottleneck is identified, the system can help evaluate better operating configurations rather than stopping at detection.
- Human-controlled — the system supports decisions without pretending that software can replace plant expertise.

8. Summary

GapTwin is built for the plant floor as it actually exists — a mix of modern, well-instrumented stations and legacy or manually operated processes — rather than the fully instrumented factory that many digital twin concepts implicitly assume.

By combining direct monitoring with confidence-scored inference, propagation awareness, vehicle-level quality risk, lead-time prediction, and constrained parameter optimization, GapTwin turns fragmented production data into a decision-ready picture of the line.

The goal is simple: detect the deviation early, understand its blast radius, identify what may be affected, estimate how much time remains, evaluate what could improve the situation, and let the plant team decide what to do.

No sensor doesn't mean no visibility.